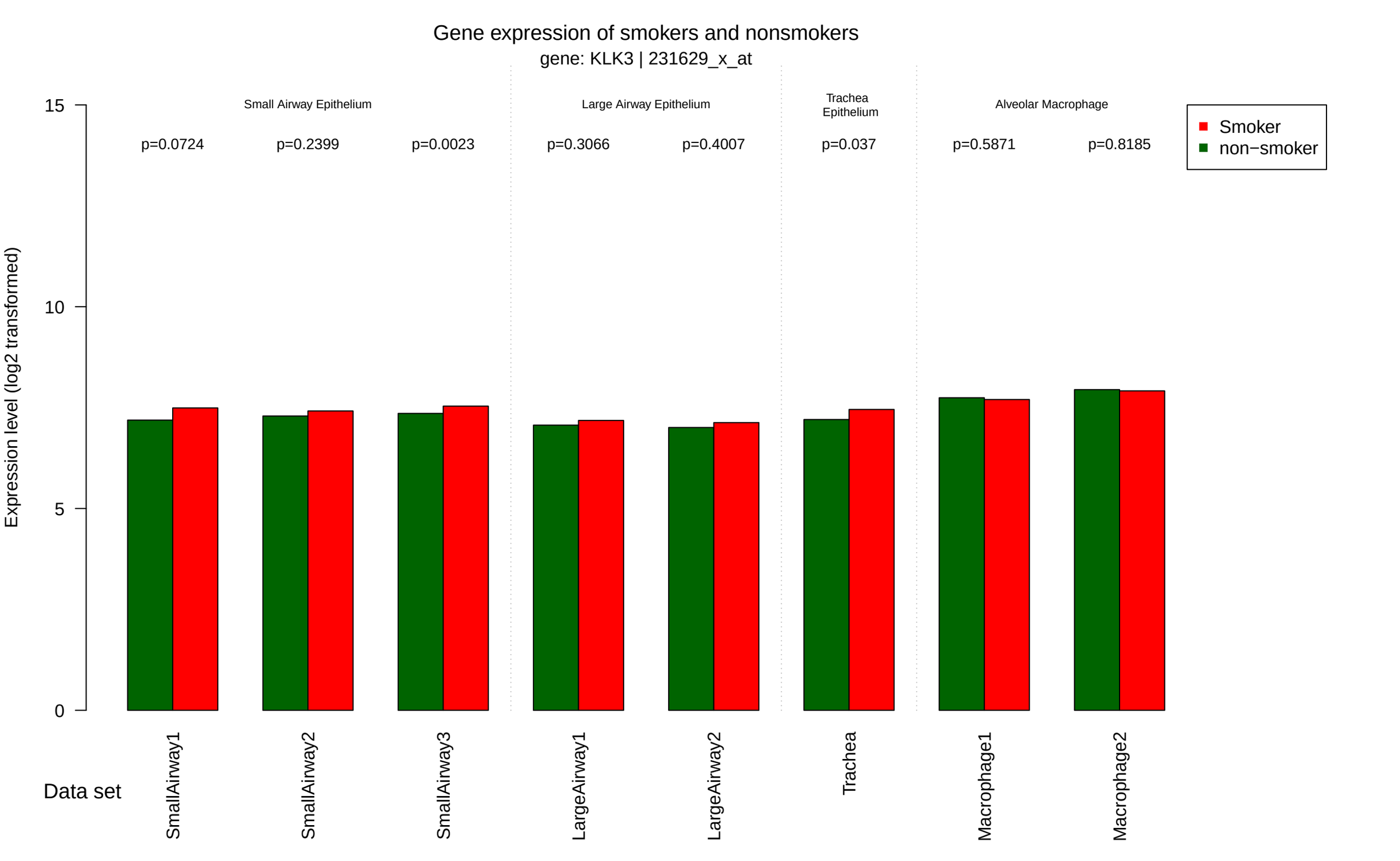
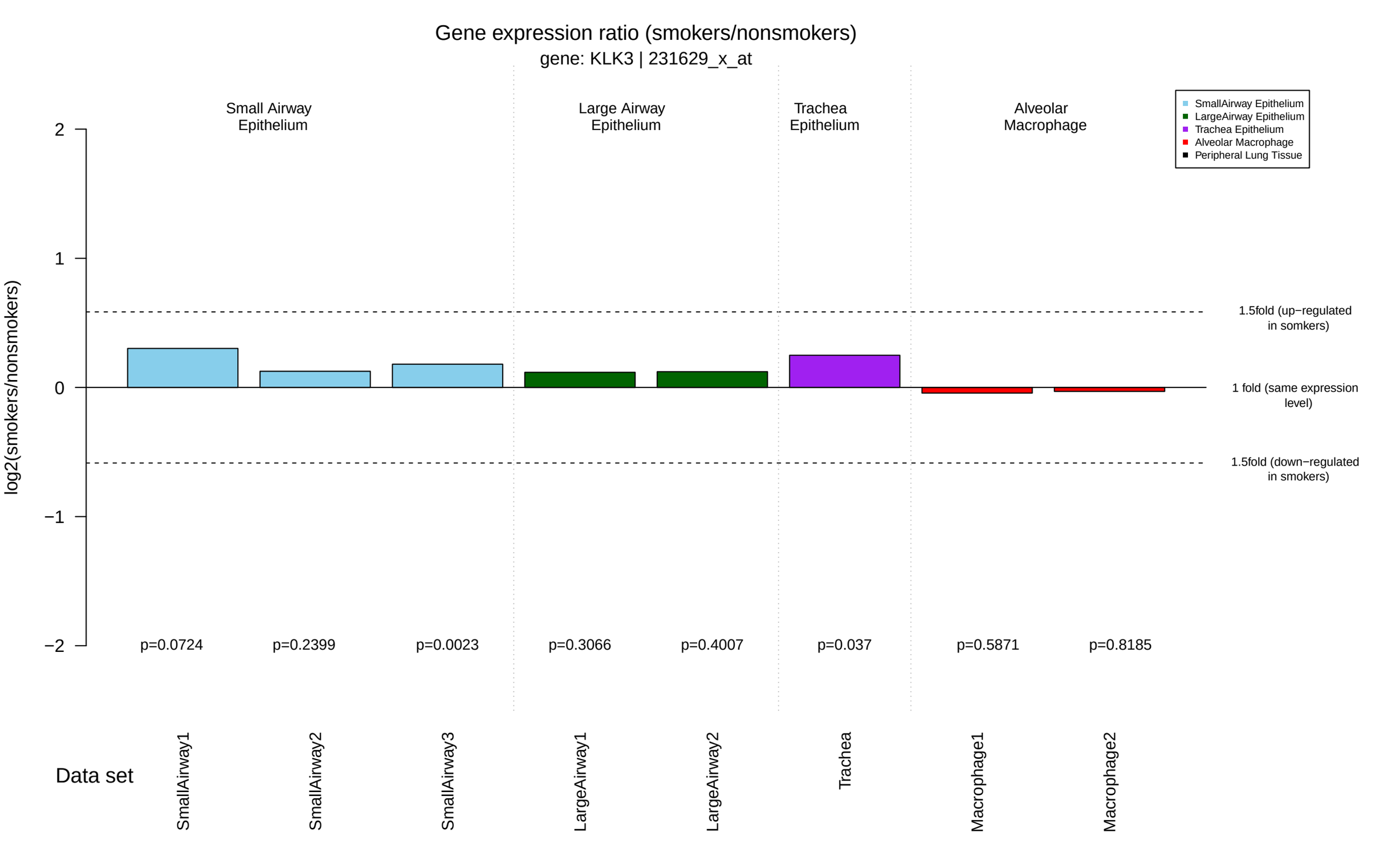
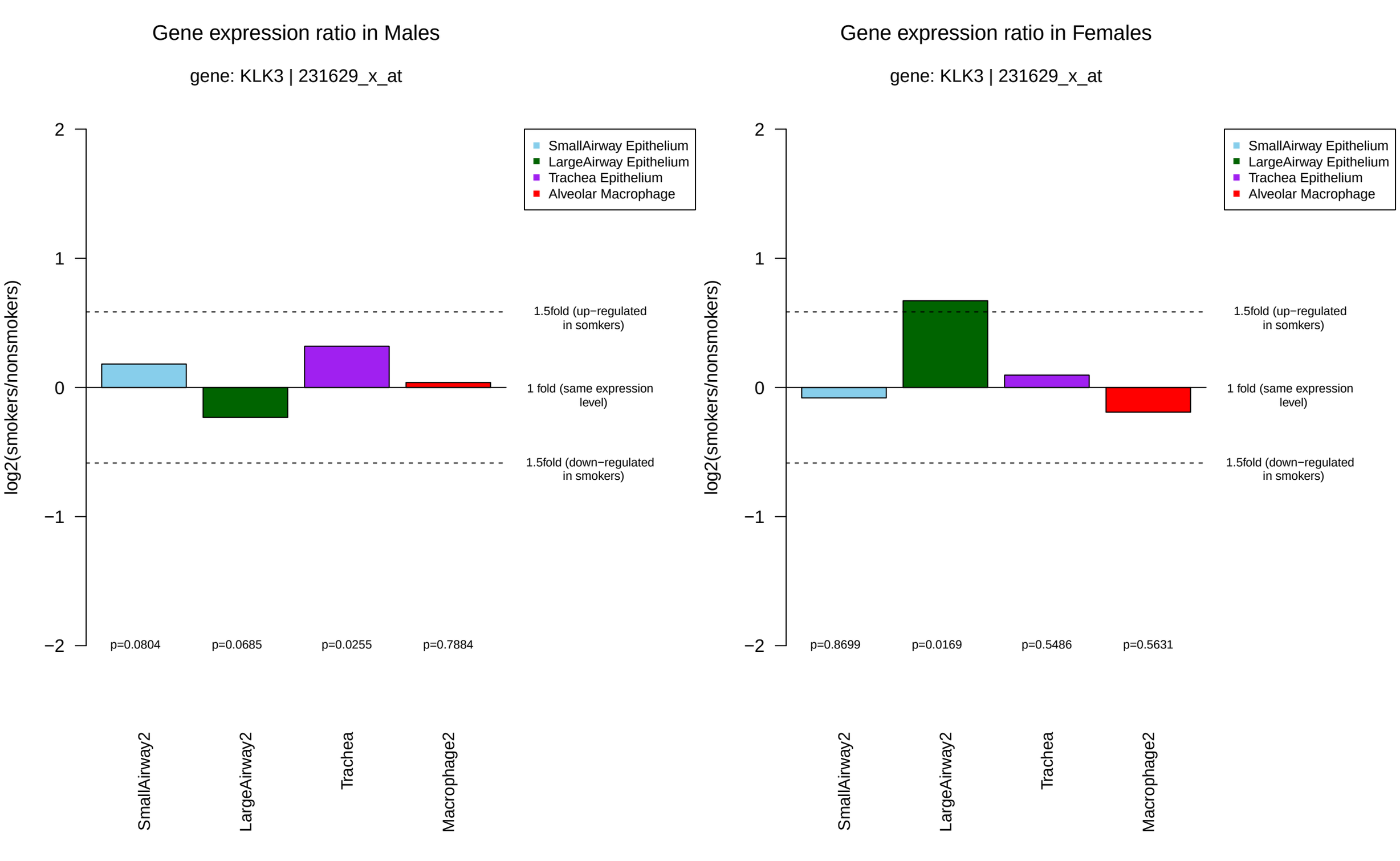
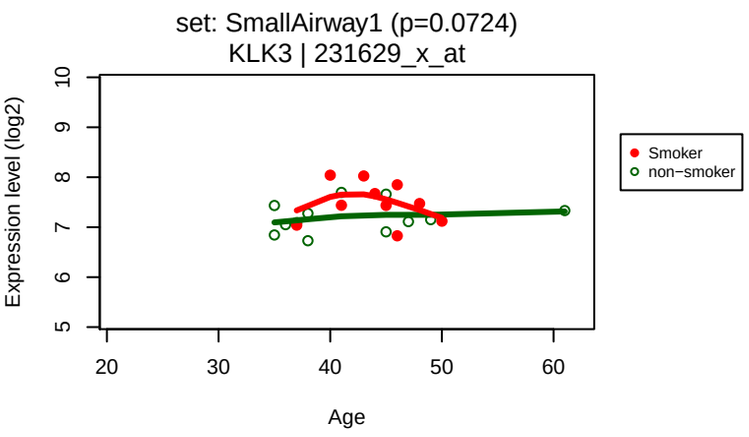
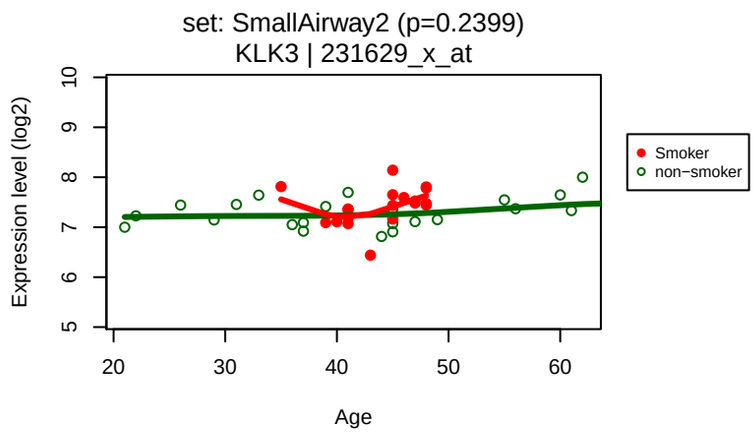
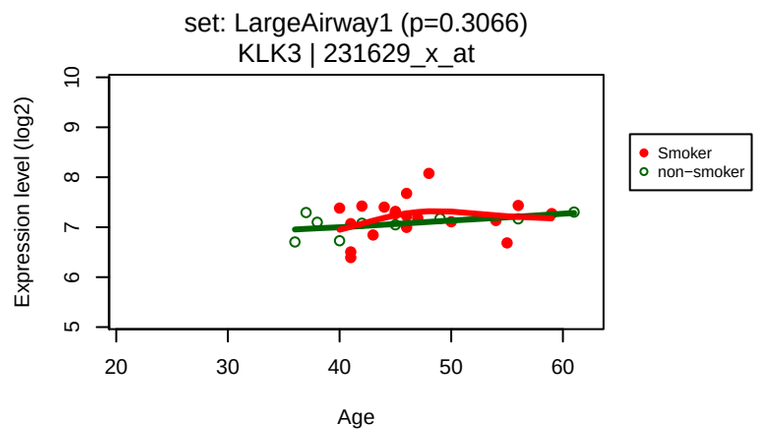

Gene expression of smokers and nonsmokers
gene: KLK3 | 231629_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0724
p=0.2399
p=0.0023
p=0.3066
p=0.4007
p=0.037
p=0.5871
p=0.8185
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

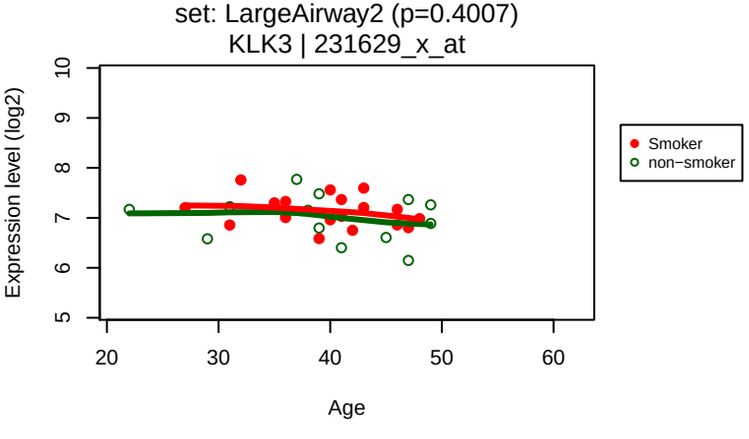

Gene expression ratio (smokers/nonsmokers)
gene: KLK3 | 231629_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0724
p=0.2399
p=0.0023
p=0.3066
p=0.4007
p=0.037
p=0.5871
p=0.8185
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

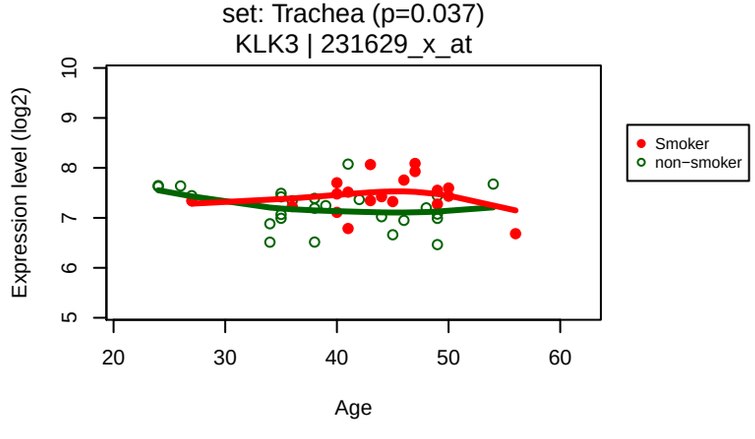

Gene expression ratio in Males
gene: KLK3 | 231629_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0804
p=0.0685
p=0.0255
p=0.7884
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK3 | 231629_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8699
p=0.0169
p=0.5486
p=0.5631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

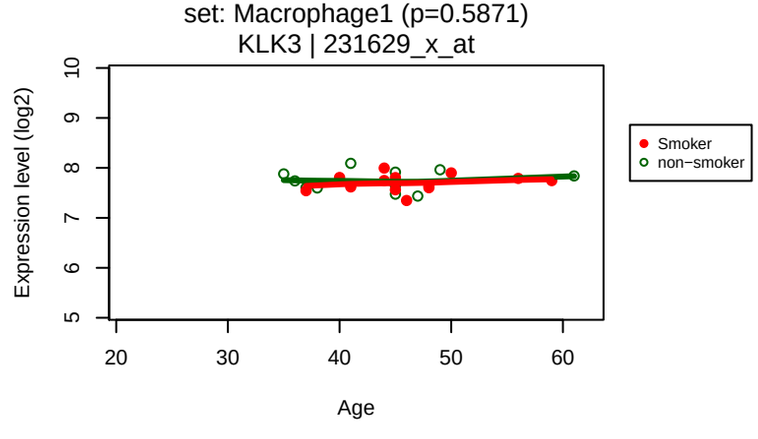

set: SmallAirway1 (p=0.0724)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker

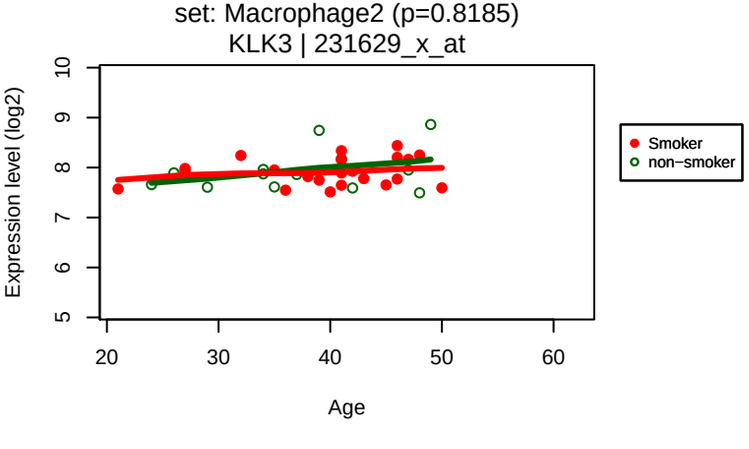

set: SmallAirway2 (p=0.2399)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker

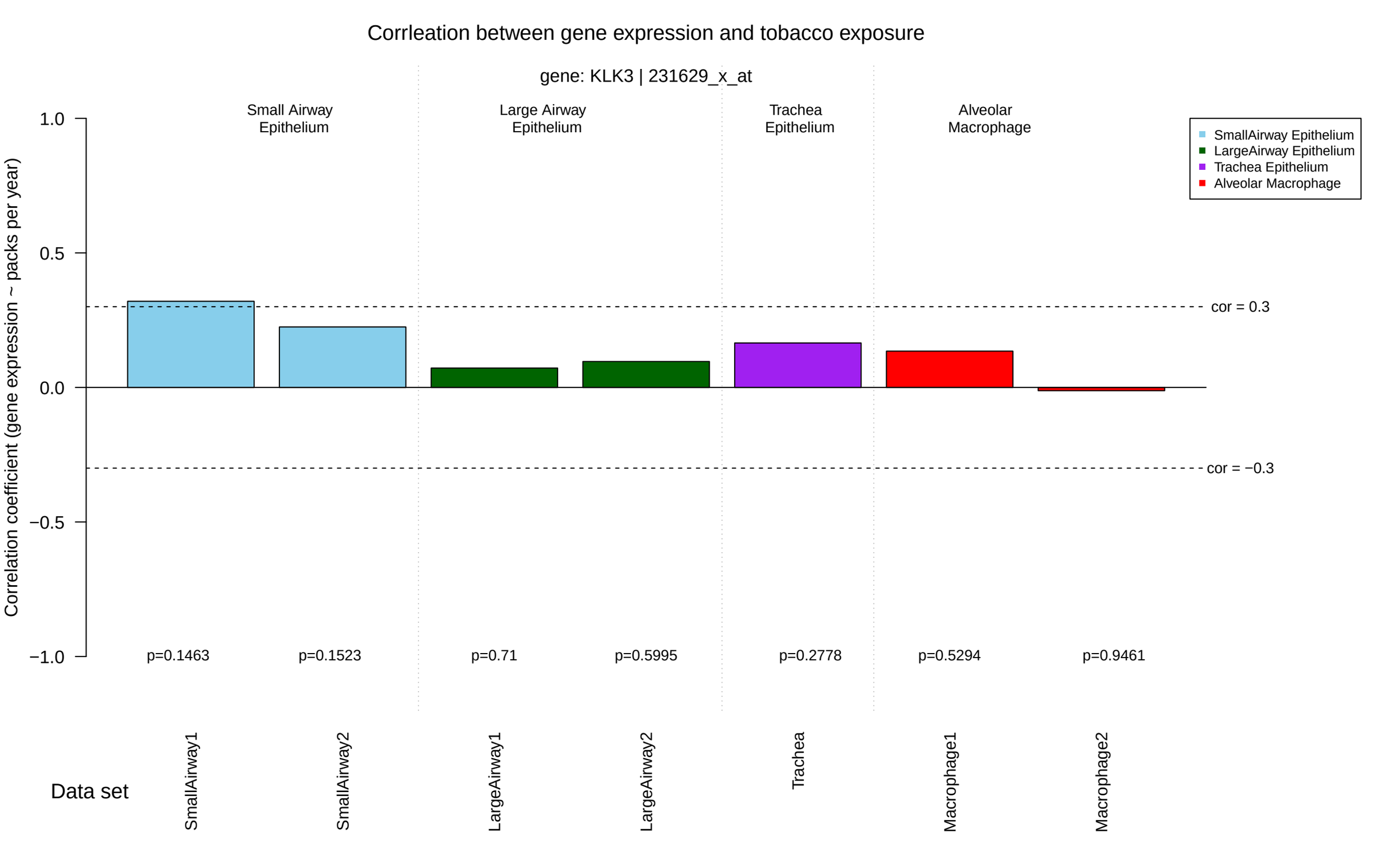

set: LargeAirway1 (p=0.3066)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4007)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.037)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5871)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8185)
KLK3 | 231629_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK3 | 231629_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1463
p=0.1523
p=0.71
p=0.5995
p=0.2778
p=0.5294
p=0.9461
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2